Beyond Captioning: A Preference-Aligned Benchmark for Fine-Grained Reasoning and Spatiotemporal Geo-World Understanding

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Abstract

Existing remote-sensing change captioning benchmarks mainly reward surface-level descriptions of visual differences, but provide limited support for evaluating whether models can perform rule-constrained semantic reasoning over observed changes, filter them under application-specific constraints, and align their outputs with geographic expert judgment. This limitation exposes a critical gap between current benchmark design and the demands of real-world Earth observation, where practical workflows require not only semantic understanding, but also higher-level reasoning and filtering. To address this gap, we introduce Geo-BC, a large-scale, preference-aligned multimodal benchmark that reframes change analysis as structured change interpretation rather than caption generation. The benchmark contains 109224 spatiotemporal samples, supplemented by 327672 reasoning clues and 382284 review comments, enabling process-level supervision beyond final-answer evaluation, as well as a curated Pro subset for challenging evaluation. In addition to benchmark instances, Geo-BC provides language-grounded reasoning trajectories, hard negative supervision for pseudo-change rejection, uncertainty-aware confidence annotations, and chosen/rejected preference pairs that support supervised fine-tuning, reward modeling, preference optimization, and fine-grained error diagnosis. We further establish baseline results across representative multimodal large models, highlighting the persistent difficulty of structured geo-world understanding. By focusing on the under-explored problem of semantically grounded change reasoning, Geo-BC provides critical data infrastructure to support the development of geospatial semantic spatiotemporal understanding. The dataset, generation code, and baseline models are publicly available at: https://github.com/kaynqi/Beyond-Captioning.

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CCS Concepts

• **Computing methodologies** → **Knowledge representation and reasoning**; **Scene understanding**; *Natural language generation*; • **Applied computing** → **Earth and atmospheric sciences**; • **General and reference** → Evaluation.

Keywords

Multimodal Reasoning, VLMs, Spatiotemporal Reasoning, Semantic Change interpretation, Benchmark Dataset, Preference-aligned Learning, Process Supervision, Earth Observation

1 Introduction

Earth observation and spatiotemporal change analysis form the foundation for understanding the dynamic evolution of the Earth, driving key applications such as urban planning, disaster response and ecological monitoring[2, 5, 6, 22]. In recent years, with the rapid advancement of multimodal large language models (MLLMs), remarkable progress has been made in remote sensing change captioning (RSCC)[4, 19, 20]. By converting visual differences between bi-temporal images into natural language, existing models have successfully bridged the gap between pixel-level changes and semantic descriptions[18, 27].

Nevertheless, the fundamental bottleneck of the current change captioning paradigm lies in its superficiality. Most existing RSCC datasets[14, 19] reward surface-level descriptions such as "new buildings appeared" or "vegetation decreased," but provide limited support for assessing whether a model can reason over spatial evidence[36], apply rule-constrained semantic filtering, reject pseudo-changes, or calibrate confidence under ambiguity. This mismatch reveals a broader limitation in current benchmark design: it measures caption ability more reliably than geographic reasoning[16, 34].

To address this gap, we introduce Geo-BC, a large-scale, preference-aligned benchmark for fine-grained reasoning and spatiotemporal

geo-world understanding. Geo-BC reframes bi-temporal remote-sensing analysis from free-form caption generation to structured change interpretation. Rather than asking a model only to state what changed, Geo-BC evaluates whether the model can localize relevant evidence, reason over spatial and semantic relations, determine whether the change is valid under a land-use taxonomy, filter out non-target variation, and express appropriate confidence when the evidence is uncertain. In this sense, the benchmark is designed not merely to test recognition, but to test decision-relevant understanding.

Geo-BC also provides supervision signals that go beyond final-answer correctness. The benchmark contains 109224 spatiotemporal samples, together with large-scale structured reasoning artifacts[11, 29, 31], review feedback, hard negatives for pseudo-change[36] rejection, uncertainty-aware confidence annotations, and chosen/rejected preference pairs. These design choices support a wide range of downstream uses, including supervised fine-tuning[24], reward modeling[7, 30], preference optimization[26], and fine-grained error diagnosis[17]. We further include a curated Pro subset to stress-test challenging cases where shallow visual difference is especially likely to be confused with semantically meaningful land-use change.

The benchmark is motivated by two observations. First, geographic change understanding requires explicit intermediate structure: robust systems should connect global scene perception, instance grounding, spatial relation modeling, semantic interpretation, future-state inference, and confidence calibration. Second, high-stakes monitoring requires not only evidence for change, but also evidence against over-interpreting non-target differences. Geo-BC operationalizes these requirements through structured reasoning trajectories and a preference-aligned supervision pipeline that makes both successful reasoning and characteristic failure modes observable.

We evaluate representative multimodal large models on Geo-BC and find that strong captioning ability does not reliably translate into strong geo-semantic reasoning. In particular, current systems remain limited when required to reject pseudo-changes, follow standards-constrained decision rules, and maintain calibrated confidence under ambiguous observations. These results highlight the difficulty of structured geographic reasoning and motivate the need for benchmarks that better reflect operational Earth observation demands.

In summary, our main contributions are as follows:

(1) **Redefine the target task:** We argue that remote sensing analysis should not be treated merely as an image captioning task. Instead, we redefine it as a structured fine-grained geographic semantic reasoning task, in which models must interpret model spatial and semantic relations and determine whether observed changes correspond to target semantic transitions under rule constraints.
(2) **A large-scale preference-aligned benchmark:** We introduce Geo-BC, a large-scale, preference-aligned benchmark with 109224 spatiotemporal samples, enriched with structured reasoning traces, hard negatives, and chosen/rejected preference pairs.
(3) **A RLAIF benchmark construction pipeline:** We develop a benchmark construction pipeline that combines a six-stage reasoning schema, positive-negative dual-track reasoning, and decoupled multi-judge evaluation, enabling both process-level supervision and fine-grained error diagnosis.
(4) **A capability gap in current multimodal models:** Our baseline experiments reveal that captioning ability in general domains does not equate to reliable structured geosemantic reasoning, especially when models are required to adhere to semantic rule constraints. As such, Geo-BC remains a challenging benchmark.

2 Methodology

To move remote sensing change analysis beyond surface-level captioning, we build a judge-supervised data generation and alignment pipeline that reframes bi-temporal remote-sensing interpretation as structured reasoning rather than free-form captioning[1, 33]. By decomposing the task into interpretable steps and evaluating them with specialized judges, the pipeline produces a preference-aligned dataset for benchmarking and downstream training[8, 35]. The overall data processing flow is shown in Fig. 1.

Due to page limitations, we summarize only the key design choices and high-level statistics here. The supplementary appendix provides detailed specifics of all the mentioned components.

2.1 Structured Multimodal Chain of Reasoning

A major limitation of conventional change captioning is that it maps a bi-temporal image pair directly to a final textual description[4, 19], leaving the intermediate reasoning process implicit. This makes it difficult to determine whether a model has grounded its interpretation in visual evidence, reasoned consistently over spatial structure, distinguished meaningful change from nuisance variation, or calibrated its confidence under ambiguous observations. To address this limitation, we represent change understanding as a Structured Multimodal Chain of Reasoning[9].

Formally, each sample is represented by an input tuple

$$(P, S, Q, I), \tag{1}$$

where P denotes the structured six-step task prompt, S denotes the land-use standard and rule constraints used during reasoning, Q denotes a JSON-form query automatically generated from the pre-change and post-change semantic maps, encoding target position and land-use semantic information, and I denotes the bi-temporal remote-sensing image pair augmented with red visual cue boxes that highlight the candidate change region[28]. Instead of directly predicting a free-form answer A in a single step, the policy model π_θ is required to generate a six-step reasoning trajectory

$$R = \{r_1, r_2, r_3, r_4, r_5, r_6\}, \tag{2}$$

followed by the final answer.

Step 1: Global perception (r_1). The first stage summarizes the overall scene and establishes a coarse semantic understanding of the bi-temporal input.

Step 2: Instance-level visual grounding (r_2). The second stage identifies salient entities, regions, and attributes, and links them to concrete visual evidence in the image pair.

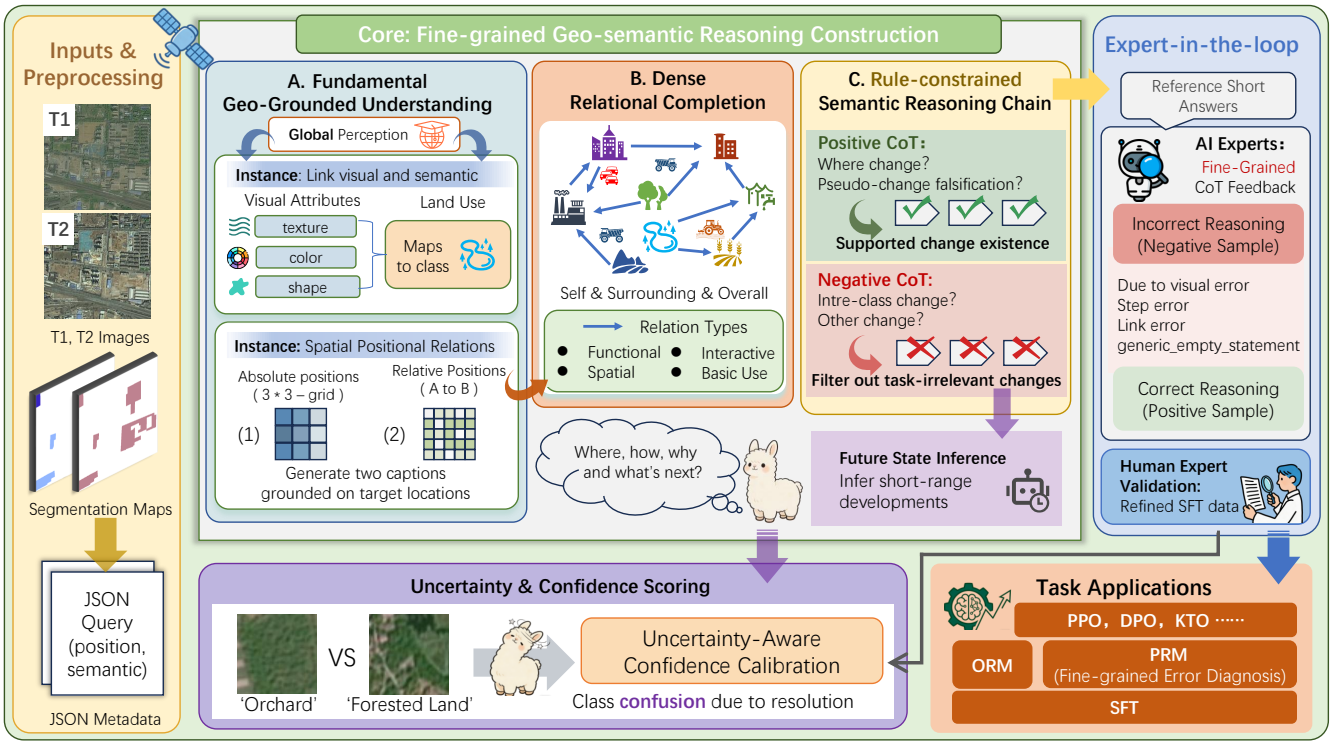

Figure 1: Overview of the proposed methodology. The illustration shows how bi-temporal remote-sensing inputs and auxiliary segmentation metadata are transformed into a structured reasoning process comprising global perception, instance grounding, dense relational completion, standards-driven semantic filtering, future-state inference, and uncertainty-aware confidence scoring, with expert-in-the-loop validation to support downstream training and benchmarking.

Step 3: Spatial relation modeling (r_3). The third stage captures the relative positions, interactions, and topological organization of grounded entities.

Step 4: Semantic reasoning (r_4). The fourth stage integrates visual evidence, grounded objects, and spatial structure to determine whether the observed differences correspond to a semantically meaningful change. This is the core reasoning stage in which the model moves from perception to interpretation.

Step 5: Future inference (r_5). The fifth stage extends the analysis from retrospective interpretation to forward-looking reasoning by predicting plausible short-term scene evolution conditioned on the current spatiotemporal state.

Step 6: Confidence correction (r_6). The final stage calibrates the output by accounting for uncertainty from limited resolution, class ambiguity, and incomplete evidence. The model then reports a confidence score under a predefined rubric, together with brief limitations or alternative explanations for ambiguous cases(The scoring criteria are detailed in the Appendix).

The data structure of the above reasoning trajectory is specifically illustrated in the left column of Fig. 2.

2.2 Positive-Negative Dual-Track Reasoning

A central challenge in remote-sensing change interpretation is that visible difference is not equivalent to operationally valid land-use change. In real monitoring scenarios, the goal is not only to determine whether two images differ, but also to decide whether the observed difference should be counted as a target event under a predefined land-use taxonomy and application protocol.

We therefore introduce a positive-negative dual-track reasoning framework that separates evidence-supported change existence from standards-constrained change validity. The positive track establishes whether a candidate change hypothesis is grounded in visual, spatial, and temporal evidence. The negative track then evaluates whether the observed difference remains within the same task-relevant land-use class or subclass and should therefore be filtered out under the current monitoring objective. In our implementation, this taxonomy follows the national land-use classification standard GB/T 21010-2017. Importantly, intra-class variation is not assumed to be unimportant in a general geographic sense; rather, such variation is treated as outside the positive target definition of this benchmark when it does not cross the semantic boundary specified by the operational taxonomy. This design better reflects high-stakes monitoring settings in which the key challenge is not merely to notice any visual change, but to determine whether the

change constitutes a semantically valid and decision-relevant land-use transition.

2.3 Decoupled Multi-Agent Criterion System

To construct a diverse candidate pool for each query, we use multiple multimodal foundation models, including **Gemini 2.5 Pro, ChatGPT 4.1, LLaVA 7B, QVQ, and Qwen 2.5 VL 7B**[3, 10, 15, 23, 25], to generate reasoning trajectories and final answers. These models cover both proprietary and open-weight systems, and are run with model-specific temperature settings to balance stability and diversity. The resulting candidates form a scalable first-pass pool, which is then filtered, ranked, and validated through structured evaluation; the output format is shown in the left panel of Fig. 2.

We evaluate these candidates using a decoupled Mixture-of-Judges (MoJ) rather than a single unified evaluator. Since long-horizon multimodal reasoning must be assessed along seven dimensions, we employ three specialized judges, each responsible for a defined subtask in the reasoning chain. To ensure consistent comparison across candidates, all judges use the same decoding configuration with a fixed temperature ($T = 0$). The evaluation criteria are illustrated in the middle panel of Fig. 2, and the step-to-judge mapping is defined as follows:

$$\begin{aligned} \text{step}_1 &\leftarrow \text{VLM}, \\ \text{step}_2 &\leftarrow \text{VLM}, \\ \text{step}_3 &\leftarrow \text{LLM}, \\ \text{step}_4 &\leftarrow \text{LLM} + \text{Rule}_{\text{ontology}}, \\ \text{step}_5 &\leftarrow \text{LLM}, \\ \text{step}_6 &\leftarrow \text{Rule}_{\text{confidence}}. \end{aligned} \tag{3}$$

Judge 1: Visual-Language Expert. Judge 1 evaluates **Step 1,2**, namely global perception and instance-level visual grounding. Its role is to verify that candidate trajectories are supported by the image pair and that referenced entities, regions, and attributes are grounded consistently in visual evidence.

Judge 2: Spatiotemporal Logic Expert. Judge 2 evaluates **Step 3,4,5**, corresponding to spatial relation modeling, causal reasoning, and future inference.

Judge 3: Standards-Compliance Judge. Judge 3 evaluates the rule-sensitive parts of the trajectory, including ontology-sensitive reasoning in **Step 4**, uncertainty-aware calibration in **Step 6**, and the **final answer**.

To assess annotation reliability, we randomly sampled 10% of the dataset and compared the AI-selected winning candidates with preference judgments provided by six human evaluators with remote-sensing research backgrounds. The human evaluation focused on whether the selected trajectories were better grounded, more semantically consistent, and more compliant with land-use standards than the rejected alternatives. We observed 94.0% agreement between the human preference judgments and the winners selected by the decoupled judging pipeline. This result suggests that the proposed multi-model annotation and multi-judge evaluation framework provides stable preference labels at scale while remaining largely consistent with expert human assessment.

2.4 Process Reward Modeling and Preference-Aligned Data Construction

Based on the decoupled MoJ system, we model the reward of the Structured Multimodal Chain of Reasoning at the *step level* and use it to construct a preference-aligned dataset. For each question and for each of the five generator models, the three judges produce structured evaluations over the relevant steps. These evaluations are then converted into scalar step scores and merged into a unified candidate record.

Let s_1 and s_2 denote the scores assigned by Judge 1, let s_3, s_4^{logic}, and s_5 denote the scores assigned by Judge 2, and let s_4^{rule} and s_6 denote the scores assigned by Judge 3. We define the process score as

$$s_{\text{proc}} = \frac{s_1 + s_2 + s_3 + s_4^{\text{logic}} + s_4^{\text{rule}} + s_5 + s_6}{7}. \tag{4}$$

This design reflects the fact that Step 4 sits at the intersection of open-ended reasoning and formal standards compliance, and therefore requires both logical and ontology-aware evaluation.

In addition to the process score, each candidate receives a hard answer score

$$s_{\text{ans}} \in \{0, 10\}, \tag{5}$$

where 10 denotes a correct final answer and 0 denotes an incorrect one. Candidates are then ranked in a two-stage manner: first by answer score, and second by process score. Formally, for the candidate set C_q associated with question q, we define

$$c_q^\star = \arg\max_{c \in C_q} \left(s_{\text{ans}}(c), s_{\text{proc}}(c) \right), \tag{6}$$

where the pair is compared in descending lexicographic order. This ranking strategy ensures that final-answer validity remains essential, while still rewarding trajectories that exhibit stronger perceptual grounding, better reasoning quality, and better compliance behavior.

The top-ranked candidate is treated as the supervised fine-tuning reference when it satisfies a minimum quality threshold. Specifically, we define the validity indicator

$$\mathbb{I}_{\text{valid}}(c_q^\star) = \mathbb{I}\left[s_{\text{ans}}(c_q^\star) = 10 \ \wedge \ s_{\text{proc}}(c_q^\star) > \tau \right], \tag{7}$$

where τ is the process-quality threshold. In our implementation, we set $\tau = 6.0$. If $\mathbb{I}_{\text{valid}}(c_q^\star) = 1$, the top-ranked candidate is retained as the chosen trajectory; otherwise, the question is marked as lacking a reliable chosen sample. All remaining candidates are retained as rejected trajectories. The resulting dataset preserves not only a single best answer, but also the full ranking structure over alternative trajectories, including detailed step scores and judge feedback.

We set $\tau = 6.0$ as a pragmatic threshold to balance quality and coverage. Because s_{proc} is an average over seven step-level scores, this value acts as a moderate filter: it removes trajectories with noticeably weak intermediate reasoning while preserving samples whose overall reasoning is sound but not perfect.

2.5 Dataset Characteristics and Statistical Analysis

The current benchmark contains 109224 spatiotemporal image samples derived from GF-1, GF-2, GF-6, and GF-7 observations, yielding 327672 inference trajectories with aligned scoring signals and

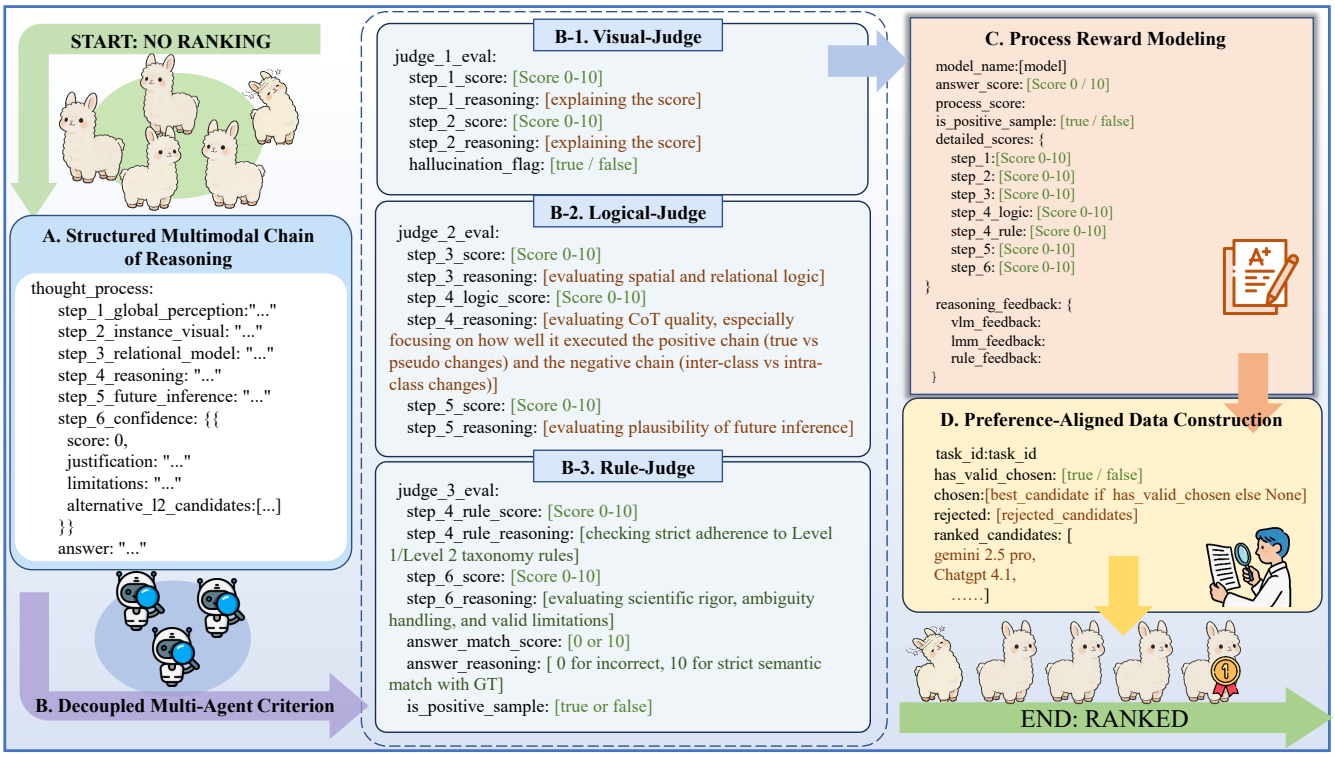

Figure 2: The specific data structure of the proposed framework. Left: Structured annotation generation. Middle: Decoupled multi-judge evaluation. Right: Ranking and preference-aligned selection.

382284 secondary evaluation statements generated by the decoupled judging framework. The overall dataset is deliberately skewed toward difficult non-target cases, with a positive-to-negative ratio of approximately 1:4, in order to reflect the realistic dominance of ambiguous or operationally irrelevant changes in practical monitoring scenarios. The present release covers 14 frequently used land-use categories under the national standard. To further support high-end benchmarking, we additionally construct a manually curated Pro subset of 39646 particularly challenging samples. This subset is designed for stress-testing fine-grained reasoning and semantic filtering, and therefore adopts a more balanced 1:1 positive-to-negative ratio.

The full dataset is divided into training, validation, and test sets in a 7:1:2 ratio. To avoid spatial leakage and inflated generalization performance, we enforce non-overlap in geographic locations across the three splits.

The end-to-end data construction pipeline consumed approximately 4.03B tokens in total, including candidate generation, multi-judge evaluation, and preference-aligned filtering/ranking; a stage-wise breakdown is provided in the supplementary material.

3 Experiments

We evaluate the utility and difficulty of Geo-BC using a diverse set of vision-language models (VLMs) spanning different architectures and domains. We benchmark model performance under progressively more challenging settings, from basic change discrimination to controlled rationale generation and template-guided structured reasoning.

3.1 Experimental Setup

3.1.1 Benchmarked Models. We benchmark six representative models: **QVQ-Max**[25], **Qwen2.5-VL-7B-Instruct**[3], **InternVL3-8B**[37], **LLaVA-OneVision**[15], **MiMo-V2-Flash**[32], and the remote-sensing foundation model **SkySense**[12]. These models cover both general-purpose MLLMs and remote-sensing-specific systems, enabling cross-architecture and cross-domain comparison on bi-temporal change interpretation.

3.1.2 Three-Stage Benchmark Suite. We design a three-stage benchmark.

The explanations of each metric are presented in Table 1. **Stage I:** Basic Change Discrimination. Given only a bi-temporal image pair, the model predicts whether a meaningful change has occurred and identifies the pre-change and post-change land-use categories. We report CDA and CC-L1, CC-L2. **Stage II:** Controlled Rationale Generation. Given a visually marked target region and its ground-truth category, the model generates an attribution rationale. We decompose the rationale into visual evidence, logical inference, and conclusion, and evaluate them using a pretrained NLI model. We report VE, LE, CE, and LHR. **Stage III:** Template-Guided Autonomous Deep Interpretation. Using the same structured reasoning template as in annotation, the model performs end-to-end interpretation from perception to geo-semantic reasoning. We evaluate outputs

Table 1: Unified comparison of six benchmarked MLLMs across the three-stage evaluation suite. Experiment I measures zero-shot basic change discrimination; Experiment II evaluates controlled zero-shot reasoning; Experiment III evaluates template-guided autonomous deep interpretation. Higher is better for all metrics except LHR.

Model	Experiment I			Experiment II				Experiment III					
	CDA ↑	CC-L1 ↑	CC-L2 ↑	VE ↑	LE ↑	CE ↑	LHR ↓	VP ↑	SL ↑	RA ↑	DD ↑	CC ↑	Overall ↑
LLaVA-OneVision[15]	50.24	0.68	0.23	41.56	67.92	40.02	0.74	2.04	1.92	1.49	1.98	2.00	9.42
InternVL3-8B[37]	51.65	10.03	0.28	37.51	61.24	11.05	16.98	1.72	1.76	1.56	1.72	1.87	8.64
Qwen2.5-VL-7B-Instruct[3]	49.97	0.42	0.25	45.80	73.05	47.47	0.31	1.80	1.70	1.52	1.72	1.60	8.33
MiMo-V2-Flash[32]	49.10	5.06	0.96	38.06	65.22	37.80	0.16	2.13	2.00	1.90	2.20	2.25	10.47
QVQ-Max[25]	50.43	11.09	0.76	43.48	72.27	40.68	0.31	2.35	2.26	1.95	2.38	2.52	11.45
SkySense[12]	78.74	2.80	0.10	29.87	59.72	29.40	1.05	1.56	1.50	1.15	1.53	1.52	7.25

Experiment I: CDA = Change Detection Accuracy; CC-L1/CC-L2 = Classification Consistency at Level-1/Level-2 categories.
Experiment II: VE = Visual Evidence; LE = Logical Evidence; CE = Conclusion Evidence; LHR = Logical Hypothesis Reliability.
Experiment III: VP = Visual Perception; SL = Spatiotemporal Logic; RA = Rule Application; DD = Domain Depth; CC = CoT Coherence.

with rubric-based scoring using ChatGPT-4o as judge, on five dimensions: VP, SL, RA, DD, and CC[21]. Each dimension is scored on a 1-5 scale.

3.1.3 Implementation Details. All experiments are conducted in a zero-shot setting. LLaVA-OneVision and InternVL are deployed locally on NVIDIA RTX 3090 GPUs, while the remaining models are accessed via official APIs. All models are prompted to output a standardized JSON format. For Stage II, rationale quality is evaluated using DeBERTa-v3-large[13] as the NLI scorer.

3.2 Results and Analysis

In Stage I, the domain-specific model SkySense[12] achieves the best change detection accuracy (CDA = 78.74%), showing a clear advantage in low-level remote-sensing perception. In contrast, general-purpose models such as LLaVA-OneVision and Qwen2.5-VL exhibit conservative zero-shot behavior and frequently miss true changes, which also lowers their CC-L1 and CC-L2 scores.

In Stage II, several general-purpose models perform better once the target region and label are provided. Qwen2.5-VL[3] achieves strong reasoning quality, with LE = 73.05 and CE = 47.47. By contrast, InternVL2.6 shows a relatively high LHR = 16.98%, indicating a tendency to generate unsupported explanations when visual grounding is weak. SkySense[12] maintains a much lower hallucination rate (LHR = 1.05%), suggesting that domain priors help suppress invalid attribution.

In Stage III, QVQ-Max[25] achieves the highest overall score 11.45, with strong performance on CC = 2.52 and VP = 2.35, indicating better multi-step reasoning coherence. In contrast, SkySense[12] performs worse in this more reasoning-intensive setting (overall = 7.25), especially on RA = 1.15 and SL = 1.50. This suggests that strong domain perception does not automatically translate into strong structured reasoning.

Our experiments demonstrate that Geo-BC reveals significant capability gaps in existing multimodal models. Domain-specialized models excel at feature extraction, while general-purpose foundation models are superior in generalized reasoning. Strong performance on general-domain description or caption-like tasks does not translate into reliable structured geospatial semantic reasoning, especially when models are required to filter non-target changes and adhere to decision rules constrained by taxonomies. So, Geo-BC remains a highly challenging benchmark.

4 Conclusion

We presented Geo-BC, a large-scale, preference-aligned benchmark that reformulates remote-sensing change analysis from surface-level captioning to structured geo-semantic reasoning, thereby addressing the lack of benchmarks that can evaluate evidence grounding, rule-constrained semantic filtering, pseudo-change rejection, and uncertainty-aware decision making in realistic Earth observation scenarios. Its design integrates structured reasoning trajectories, dual-track positive/negative supervision, multi-judge evaluation, and preference-aligned data construction, providing a richer and more practical training and evaluation framework than existing RSCC resources. With 109224 samples, 327672 reasoning trajectories, and a curated Pro subset, Geo-BC not only fills an important research gap in fine-grained spatiotemporal geo-world understanding, but also offers strong innovation and clear practical value for downstream tasks such as supervised fine-tuning, reward modeling, preference optimization, and error diagnosis. Moreover, the benchmark results show that current multimodal models still struggle to achieve reliable structured reasoning under geographic constraints, underscoring both the challenge and the necessity of this dataset for advancing robust and operational Earth observation intelligence.

5 Limitations

Geo-BC is currently built under a specific land-use standard and mainly based on samples from North China, which may limit its generalization to broader Earth-observation settings. In future work, we will expand the dataset to more regions and more diverse land-use conditions. Although the current release does not yet include satellite data from other countries, the proposed RLAIF-based annotation and review framework is general and reusable, and we plan to open-source the full toolkit and pipeline to support more universal benchmark construction.

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